

Assignment 1

Que. Expand the van der Waals equation of state as a cubic equation of volume. At the critical point, the three roots of this equation must collapse to a single root, i.e., $(V-V_C)^3 = 0$. Compare these two equations term-by-term and evaluate the critical volume, pressure, and temperature in terms of the constants a , b , and R . Now, define the reduced volume/pressure/temperature as the volume/pressure/temperature divided by the critical volume/pressure/temperature and re-write the van der Waals equation in terms of reduced parameters. What is the importance of this equation?

E1C.9(a) A certain gas obeys the van der Waals equation with $a = 0.50 \text{ m}^6 \text{ Pa mol}^{-2}$. Its molar volume is found to be $5.00 \times 10^{-4} \text{ m}^3 \text{ mol}^{-1}$ at 273 K and 3.0 MPa. From this information calculate the van der Waals constant b . What is the compression factor for this gas at the prevailing temperature and pressure?

E2A.4(a) A sample consisting of 1.00 mol Ar is expanded isothermally at 20 °C from 10.0 dm³ to 30.0 dm³ (i) reversibly, (ii) against a constant external pressure equal to the final pressure of the gas, and (iii) freely (against zero external pressure). For the three processes calculate q , w , and ΔU .

E2A.5(a) A sample consisting of 1.00 mol of perfect gas atoms, for which $C_{V,m} = \frac{3}{2}R$, initially at $p_1 = 1.00 \text{ atm}$ and $T_1 = 300 \text{ K}$, is heated reversibly to 400 K at constant volume. Calculate the final pressure, ΔU , q , and w .

E2B.4(a) When 3.0 mol O₂ is heated at a constant pressure of 3.25 atm, its temperature increases from 260 K to 285 K. Given that the molar heat capacity of O₂ at constant pressure is $29.4 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate q , ΔH , and ΔU .

P2D.2 Starting from the expression $C_p - C_v = T(\partial p/\partial T)_v(\partial V/\partial T)_p$, use the appropriate relations between partial derivatives (*The chemist's toolkit 9* in Topic 2A) to show that

$$C_p - C_v = \frac{T(\partial V/\partial T)_p^2}{(\partial V/\partial p)_T}$$

Use this expression to evaluate $C_p - C_v$ for a perfect gas.

2.20 Show that the following functions have exact differentials: (a) $x^2y + 3y^2$, (b) $x \cos xy$, (c) x^3y^2 , (d) $t(t + e^x) + s$.

E2E.1(a) Use the equipartition principle to estimate the values of $\gamma = C_p/C_v$ for gaseous ammonia and methane. Do this calculation with and without the vibrational contribution to the energy. Which is closer to the experimental value at 25 °C?

The experimental values of γ for ammonia and methane are the same, $\gamma = 1.31$.

I2.5 As shown in *The chemist's toolkit 9* in Topic 2A, it is a property of partial derivatives that

$$\left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)_y \right)_x = \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)_x \right)_y$$

Use this property and eqn 2A.14 to write an expression for $(\partial C_v/\partial V)_T$ as a second derivative of U and find its relation to $(\partial U/\partial V)_T$. Then show that $(\partial C_v/\partial V)_T = 0$ for a perfect gas.

E3A.2(a) Consider a process in which 100 kJ of energy is transferred reversibly and isothermally as heat to a large block of copper. Calculate the change in entropy of the block if the process takes place at (i) 0 °C, (ii) 50 °C.

E3A.5(a) In a certain ideal heat engine, 10.00 kJ of heat is withdrawn from the hot source at 273 K and 3.00 kJ of work is generated. What is the temperature of the cold sink?

E3B.7(a) Calculate the change in entropy of the system when 10.0 g of ice at -10.0°C is converted into water vapour at 115.0°C and at a constant pressure of 1 bar. The molar constant-pressure heat capacities are: $C_{p,m}(\text{H}_2\text{O}(\text{s})) = 37.6 \text{ J K}^{-1} \text{ mol}^{-1}$; $C_{p,m}(\text{H}_2\text{O}(\text{l})) = 75.3 \text{ J K}^{-1} \text{ mol}^{-1}$; and $C_{p,m}(\text{H}_2\text{O}(\text{g})) = 33.6 \text{ J K}^{-1} \text{ mol}^{-1}$. The standard enthalpy of vaporization of $\text{H}_2\text{O}(\text{l})$ is 40.7 kJ mol^{-1} , and the standard enthalpy of fusion of $\text{H}_2\text{O}(\text{l})$ is 6.01 kJ mol^{-1} , both at the relevant transition temperatures.

E3C.1(a) At 4.2 K the heat capacity of $\text{Ag}(\text{s})$ is $0.0145 \text{ J K}^{-1} \text{ mol}^{-1}$. Assuming that the Debye law applies, determine $S_m(4.2 \text{ K}) - S_m(0)$ for silver.

Question:

Starting from the definitions of the thermodynamic potentials $H = U + pV$, $A = U - TS$, and $G = H - TS$, derive the remaining three Maxwell relations. Form the exact differentials of H , A , and G , substitute $dU = TdS - pdV$, and obtain the corresponding Maxwell relations. Hence write down the complete set of Maxwell relations.